A REVIEW OF GREEN'S, STOKES'S, AND THE DIVERGENCE THEOREM

ABSTRACT. The goal of this review is to summarize some frequently used theorems in vector calculus, along with examples and practice problems. The intended audience is students currently taking a multivariable calculus course who want to deepen their understanding of the results and their presentation. However, this should not be their first introduction to these results. Audiences who have previously seen these theorems and want a concise refresher may also find this document beneficial. At the end, we include an application of Green's Theorem used in distribution theory that motivates this exposition, along with its role in deriving the Brown measure of non-normal operators.

Sec.	**Main Topics**	**Description and Sources**
(1)	Introduction	Unifying viewpoint of the Fundamental Theorem of Calculus across dimensions.
(2)	Summary of Main Theorems	Overview of the circulation/curl and flux/divergence theorems in $\mathbb{R}^2$ and $\mathbb{R}^3$.
(8)	Exercises	Practice problems identifying which integral theorem to apply, with hints and answers.
(9.2)	An Application of Green's Theorem	We show that $\frac{1}{2\pi}\log(r)$ is the fundamental solution to the 2D distributional Laplacian using Green's Theorem, motivating Brown measure.

1. Introduction

In single-variable calculus, the **Fundamental Theorem of Calculus (FTC)** states that if $f : [a, b] \to \mathbb{R}$ is continuously differentiable, then

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

Conceptually, this says that the integral of a *derivative* of a function over an interior region (the interval $[a, b]$) is completely determined by the values of the original function on the *oriented boundary* of that region (the endpoints $\partial[a, b] = \{a, b\}$).

The main theorems of vector calculus—the Fundamental Theorem for Line Integrals, Green's Theorem (in both circulation and flux forms), Stokes's Theorem, and the Divergence Theorem (Gauss's Theorem)—are all higher-dimensional realizations of this exact same principle:

$$\int_\Omega (\text{Derivative of } \vec{F}) = \int_{\partial\Omega} \vec{F}.$$

Depending on the dimension of the domain Ω and the type of vector derivative we take (gradient ∇f, 2D/3D curl $\nabla \times \vec{F}$, or divergence $\nabla \cdot \vec{F}$), we obtain two natural families of theorems:

(1) **Circulation / Curl Theorems (Tangential Form):** Relate the integral of the *curl* of a vector field over a 2D region or surface to the *circulation* (tangential line integral) of the vector field around its closed boundary curve.
(2) **Flux / Divergence Theorems (Normal Form):** Relate the integral of the *divergence* of a vector field over a 2D region or 3D solid to the *outward flux* of the vector field across its closed boundary curve or surface.

The goal of these notes is two-fold: first, to help you quickly recognize which theorem applies to a given integral by looking at the dimension of the domain, whether the boundary is closed, and whether you are integrating a tangential component ($\vec{F} \cdot \hat{T}$) or a normal component ($\vec{F} \cdot \hat{n}$); and second, to demonstrate how these classical calculus tools serve as the foundation for modern analysis, specifically distribution theory and the Brown measure of operators.

2. Summary of Main Theorems

(1) (3.1) **Green's Theorem (Circulation Form / 2D-Curl Form):**

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

(2) (4.1) **Generalized Green's Theorem (Regions with Holes):**

$$\oint_{\partial D} \vec{F} \cdot d\vec{r} = \oint_{C_{\text{outer}}} \vec{F} \cdot d\vec{r} + \sum_{k=1}^{m} \oint_{C_{\text{inner},k}} \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

(3) (5.1) **Stokes's Theorem (3D Curl Form):**

$$\oint_{\partial S} \vec{F} \cdot \hat{T}\, ds = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS.$$

(4) (6.1) **Green's Theorem (Flux Form / 2D-Divergence Form):**

$$\oint_{\partial D} \vec{F} \cdot \hat{n}\, ds = \iint_D (\nabla \cdot \vec{F})\, dA = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dA.$$

(5) (7.1) **The Divergence Theorem (3D Flux Form):**

$$\oiint_{\partial E} \vec{F} \cdot \hat{n}\, dS = \iiint_E (\nabla \cdot \vec{F})\, dV.$$

3. Green's Theorem (Circulation Form / 2D-Curl Form)

Theorem 3.1. *Let $D \subseteq \mathbb{R}^2$ be a bounded, simply connected plane region whose boundary ∂D is a simple, closed, piecewise-smooth curve oriented positively (counterclockwise). Suppose $\vec{F}(x,y) = \langle P(x,y), Q(x,y)\rangle$ is a vector field whose component functions have continuous first-order partial derivatives on an open region containing D. Then*

$$\oint_{\partial D} \vec{F}\cdot\hat{T}\,ds = \oint_{\partial D} (P\,dx + Q\,dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA,$$

where $\hat{T}$ is the unit tangent vector to ∂D pointing in the direction of positive orientation.

Remark 3.2. **Positive Orientation:** A boundary curve ∂D is *positively oriented* if, as you walk along ∂D in the direction of $\hat{T}$, the region D is always to your **left**. For a simple closed curve enclosing a region D, positive orientation is **counterclockwise**. If a curve C is oriented *clockwise*, we must introduce a negative sign:

$$\oint_{C_{\text{clockwise}}} \vec{F}\cdot d\vec{r} = -\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Remark 3.3. **Why "2D-Curl Form"?** If we embed $\vec{F}(x,y) = \langle P(x,y), Q(x,y), 0\rangle$ into $\mathbb{R}^3$, then its 3D curl is

$$\nabla\times\vec{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ P(x,y) & Q(x,y) & 0 \end{vmatrix} = \left\langle 0,\, 0,\, \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right\rangle.$$

Taking the upward unit normal $\hat{n} = \mathbf{k} = \langle 0,0,1\rangle$ to the xy-plane gives $(\nabla\times\vec{F})\cdot\mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$, called the **scalar (2D) curl** of $\vec{F}$. Thus, Green's Circulation Theorem is the planar special case of Stokes's Theorem (5.1).

4. Generalized Green's Theorem (Regions with Holes)

What happens if the region $D \subseteq \mathbb{R}^2$ is *not* simply connected (for instance, an annulus $A_{r_0,R} = \{(x,y) \in \mathbb{R}^2 : r_0 \leq \sqrt{x^2+y^2} \leq R\}$ with a hole around the origin)? By slicing D into simply connected subregions using auxiliary cuts whose line integrals cancel in opposite directions, Green's Theorem extends to multiply connected domains.

Theorem 4.1. *Let $D \subseteq \mathbb{R}^2$ be a bounded region with piecewise-smooth boundary $\partial D = C_0 \cup C_1 \cup \cdots \cup C_m$, where C_0 is the outer boundary curve oriented **counterclockwise** and $C_1, \ldots, C_m$ are the inner boundary curves (around the holes) oriented **clockwise** (so that D always lies to the left as one traverses ∂D). If $\vec{F} = \langle P, Q \rangle$ has continuous first-order partial derivatives on an open region containing D, then:*

(a) Circulation Form:

$$\oint_{\partial D} \vec{F} \cdot \hat{T}\, ds = \oint_{C_0} \vec{F} \cdot d\vec{r} + \sum_{k=1}^{m} \oint_{C_k} \vec{F} \cdot d\vec{r} = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA.$$

(b) Flux / Divergence Form:

$$\oint_{\partial D} \vec{F} \cdot \hat{n}\, ds = \oint_{C_0} \vec{F} \cdot \hat{n}\, ds + \sum_{k=1}^{m} \oint_{C_k} \vec{F} \cdot \hat{n}\, ds = \iint_D (\nabla \cdot \vec{F})\, dA,$$

*where $\hat{n}$ is the unit normal vector pointing **outward from** D (meaning away from the interior of D: radially outward on C_0, and inward into each hole on $C_1, \ldots, C_m$).*

Remark 4.2. **Key Consequence for Singular Vector Fields:** Suppose $\vec{F}$ has a singularity at a point $p_0 \in \mathbb{R}^2$, Automatic application of standard Green's Theorem on a disk containing p_0 is invalid because $\vec{F}$ is not C^1 at p_0. However, by excising a small disk $B_{r_0}(p_0)$ around the singularity, we can legally apply Generalized Green's Theorem (4.1) on the punctured region $D \setminus B_{r_0}(p_0)$ and then take the limit as $r_0 \to 0^+$. This is the exact strategy we use in Section 9 to prove Lemma 9.2.

5. Stokes's Theorem

Theorem 5.1. *Let S be an oriented, piecewise-smooth surface in $\mathbb{R}^3$, with ∂S being its positively oriented, simple, closed, piecewise-smooth boundary curve. Suppose $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ is a vector field whose components have continuous first-order partial derivatives on an open region containing S. Then*

$$\oint_{\partial S} \vec{F} \cdot \hat{T}\, ds = \iint_S (\nabla \times \vec{F}) \cdot \hat{n}\, dS,$$

where $\hat{\mathbf{n}}$ is the unit normal vector to S compatibility-oriented with ∂S via the right-hand rule, and $\hat{\mathbf{T}}$ is the unit tangent vector to the boundary curve ∂S.

Remark 5.2. That is, the circulation of $\vec{\mathbf{F}}$ around the boundary curve ∂S is equal to the flux of the curl of $\vec{\mathbf{F}}$ through the surface S.

Remark 5.3. On the right-hand side we compute a surface integral, and on the left-hand side a line integral. Recall the shorthand vector differential notations $d\vec{\mathbf{S}} = \hat{\mathbf{n}}\, dS$ and $d\vec{\mathbf{r}} = \hat{\mathbf{T}}\, ds$. Furthermore, if two different oriented surfaces S_1 and S_2 share the same oriented boundary curve $\partial S_1 = \partial S_2$, Stokes's Theorem guarantees that $\iint_{S_1} (\nabla \times \vec{F}) \cdot \hat{n}\, dS = \iint_{S_2} (\nabla \times \vec{F}) \cdot \hat{n}\, dS$ ("surface independence" of curl flux).

6. Green's Theorem (Flux / Divergence Form)

Recall that a subset of the plane $D \subseteq \mathbb{R}^2$ is *simply connected* if it is path-connected and has no holes passing through it (i.e., any simple closed curve in D can be continuously shrunk to a single point without exiting D).

Theorem 6.1. *Let $D \subseteq \mathbb{R}^2$ be a bounded, simply connected region in the plane with a piecewise-smooth boundary curve ∂D positively oriented with respect to D. Suppose $\vec{F} = \langle P, Q\rangle : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ is a vector field whose component functions have continuous first-order partial derivatives on an open region U containing D. Then*

$$\oint_{\partial D} \vec{F} \cdot \hat{n}\, ds = \iint_D (\nabla \cdot \vec{F})\, dA = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}\right) dA,$$

where $\hat{n}$ denotes the outward-pointing unit normal vector to the boundary curve ∂D at each point.

The flux form of Green's Theorem follows immediately from the circulation form (3.1) after observing that for a vector field $\vec{F} = \langle P, Q\rangle$, unit tangent $\hat{T}$, and outward unit normal $\hat{n}$ at each point of the positively oriented curve ∂D, we have

$$\vec{F}(x, y) \cdot \hat{n}(x, y) = \langle -Q(x, y), P(x, y)\rangle \cdot \hat{T}(x, y),$$

so one simply applies the circulation form of Green's Theorem to the auxiliary vector field $\vec{G} = \langle -Q, P\rangle$.

Indeed, if $\vec{r}(t) = \langle x(t), y(t)\rangle$ $(a \le t \le b)$ is a positively oriented (counterclockwise) parameterization of ∂D, recall that the unit tangent vector at $(x_0, y_0) = (x(t_0), y(t_0))$ is

$$\hat{T}(x_0, y_0) = \frac{\vec{r}\,'(t_0)}{\|\vec{r}\,'(t_0)\|} = \frac{1}{\|\vec{r}\,'(t_0)\|}\langle x'(t_0), y'(t_0)\rangle.$$

Rotating $\hat{T}$ clockwise by 90° (so that it points outward to the right of the counterclockwise path) yields the outward unit normal vector by interchanging components and negating the second component:

$$\hat{n}(x_0, y_0) = \frac{1}{\|\vec{r}\,'(t_0)\|}\langle y'(t_0), -x'(t_0)\rangle.$$

Since $ds = \|\vec{r}\,'(t)\|\, dt$, we obtain $\vec{F} \cdot \hat{n}\, ds = P\, dy - Q\, dx$.

Remark 6.2. This is referred to as **Green's Flux Form** because the line integral of $\vec{F} \cdot \hat{n}$ around the closed curve ∂D measures the net *outward flux* of $\vec{F}$ across ∂D. It is also called the **2D Divergence Theorem** because $\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ is the divergence of $\vec{F}$.

7. The Divergence Theorem

Theorem 7.1. *Let $E \subseteq \mathbb{R}^3$ be a simple solid region (bounded and closed) whose boundary surface ∂E is a closed, piecewise-smooth surface oriented outward. Then*

$$\oiint_{\partial E} \vec{F} \cdot \hat{n}\, dS = \iiint_E (\nabla \cdot \vec{F})\, dV$$

for any vector field $\vec{F} : U \subseteq \mathbb{R}^3 \to \mathbb{R}^3$ whose component functions have continuous first-order partial derivatives on an open region U containing E, where $\hat{n}$ denotes the outward-pointing unit normal to ∂E at each point.

Remark 7.2. The left-hand side computes the net outward **flux** of the vector field $\vec{F}$ across the closed boundary surface ∂E. That is, it measures the net rate at which fluid (modeled by $\vec{F}$) exits vs. enters the closed surface ∂E, equating it to the total source/sink density $(\nabla \cdot \vec{F})$ integrated throughout the interior solid E.

8. Exercises

The purpose of the exercises below is to practice identifying which vector calculus theorem to apply.

Problem 1

Let $\mathbf{F} = \langle e^{xy}, e^{xz}, x^2 z \rangle$ be a vector field, and let S be the surface consisting of the **right half** (i.e., where $y \geq 0$) of the ellipsoid

$$2x^2 + y^2 + 4z^2 = 4,$$

which lies to the **right of the xz-plane**, oriented in the direction of the **positive y-axis**.

Compute the surface integral of the curl of $\mathbf{F}$ over S:

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS,$$

where $\mathbf{n}$ is the unit normal vector pointing in the direction of the positive y-axis.

Problem 2

Compute the work done by the force field $\mathbf{F} = \langle x(x+y),\, xy^2 \rangle$ in moving a particle along the following closed path C:

- From the origin $(0,0)$ along the x-axis to the point $A = (1,0)$,
- Then along the line segment from $A = (1,0)$ to $B = (0,1)$,
- And finally back to the origin $(0,0)$ along the y-axis.

That is, compute the line integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

around the boundary C of the enclosed triangular region.

Problem 3

Let $\mathbf{F}(x, y) = \langle y - \cos y,\ x \sin y \rangle$ be a vector field, and let C be the circle

$$(x - 3)^2 + (y + 4)^2 = 4$$

oriented in the **clockwise direction**.

(a) Evaluate the line integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r}.$$

Problem 4

Let $\mathbf{F}(x, y) = \langle x^2y^3,\ x^3y^2 \rangle$ be a vector field.

(a) Show that $\mathbf{F}$ is a conservative vector field. That is, find a scalar potential function $f(x, y)$ such that

$$\nabla f = \mathbf{F}.$$

(b) Let C be the curve given by the parametric vector function

$$\mathbf{r}(t) = \langle t^3 - 2t,\ t^3 + 2t \rangle, \quad \text{for } t \in [0, 1].$$

Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{r}.$$

Problem 5

Consider the vector field

$$\mathbf{F}(x, y, z) = \langle \cos z + xy^2,\ xe^{-z},\ \sin y + x^2 z \rangle.$$

Let S be the closed boundary surface (oriented outward) of the solid E bounded below by the paraboloid

$$z = x^2 + y^2$$

and capped from above by the plane $z = 4$.

(a) Compute the total outward flux of $\mathbf{F}$ across the closed surface S:

$$\oiint_S \mathbf{F} \cdot \mathbf{n}\, dS.$$

Hints and Answers

Problem 1.

- **Hint:** Apply Stokes's Theorem! The boundary curve ∂S lies in the xz-plane ($y = 0$), where it forms the ellipse $2x^2 + 4z^2 = 4$. Alternatively, replace S with the flat elliptical disk $D_{xz} = \{(x, 0, z) : 2x^2 + 4z^2 \le 4\}$ in the $y = 0$ plane oriented with unit normal $\mathbf{j} = \langle 0, 1, 0\rangle$. Since $(\nabla \times \mathbf{F}) \cdot \mathbf{j} = \frac{\partial}{\partial z}(e^{xy}) - \frac{\partial}{\partial x}(x^2 z) = -2xz$ is an odd function of x (and z) over the symmetric ellipse D_{xz}, its double integral vanishes.
- **Answer:** $\boxed{0}$

Problem 2.

- **Hint:** The path C is a simple closed curve oriented counterclockwise around the triangle $D = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le 1 - x\}$. Green's Theorem converts the circulation integral into the double integral $\iint_D (y^2 - x)\, dA$.
- **Answer:** $\boxed{-\dfrac{1}{12}}$

Problem 3.

- **Hint:** The enclosed region D is a disk of radius $r = 2$. Here $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = \sin y - (1 + \sin y) = -1$. Apply Green's Theorem and include an extra minus sign to account for the **clockwise** orientation of C: $-\iint_D (-1)\, dA = \text{Area}(D)$.
- **Answer:** $\boxed{4\pi}$

Problem 4.

- **Hint (a):** Verify that $\frac{\partial P}{\partial y} = 3x^2 y^2 = \frac{\partial Q}{\partial x}$ on $\mathbb{R}^2$. Integrate $F_1 = x^2 y^3$ with respect to x and match with F_2.
- **Hint (b):** Use the Fundamental Theorem for Line Integrals: $\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) = f(-1, 3) - f(0, 0)$.
- **Answer:**

$$f(x, y) = \frac{1}{3} x^3 y^3 + C, \qquad \int_C \mathbf{F} \cdot d\mathbf{r} = \boxed{-9}$$

Problem 5.

- **Hint:** Since S is a closed surface, apply the Divergence Theorem:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(\cos z + xy^2) + \frac{\partial}{\partial y}(xe^{-z}) + \frac{\partial}{\partial z}(\sin y + x^2 z) = y^2 + 0 + x^2 = x^2 + y^2.$$

  Then evaluate $\iiint_E (x^2 + y^2)\, dV$ in cylindrical coordinates over $0 \le \theta \le 2\pi$, $0 \le r \le 2$, and $r^2 \le z \le 4$.
- **Answer:** $\boxed{\dfrac{32\pi}{3}}$

9. An Application of Green's Theorem (Flux Form)

The main goal of this section is to prove Lemma 9.2 and explain its significance in distribution theory and operator algebras. Ideas here are not mine and is taken from [1] see section on Brown measure.

Background: Distribution Theory and Brown Measure. In classical analysis, many physically and mathematically essential objects—such as the **Dirac delta point mass** δ_0 at the origin, or functions with singularities—are not classically differentiable. **Distribution theory** (developed by Laurent Schwartz) resolves this by shifting derivatives onto smooth **test functions**.

- **Test Functions ($C_c^\infty(\mathbb{R}^2)$):** A function $\psi : \mathbb{R}^2 \to \mathbb{R}$ (or $\mathbb{C}$) belongs to $C_c^\infty(\mathbb{R}^2)$ if it is infinitely differentiable (smooth) and has *compact support*, meaning $\operatorname{supp}(\psi) := \overline{\{(x,y) \in \mathbb{R}^2 : \psi(x,y) \neq 0\}}$ is a closed and bounded subset of $\mathbb{R}^2$.
- **Distributions and Distributional Derivatives:** Any locally integrable function $u \in L^1_{\text{loc}}(\mathbb{R}^2)$ defines a continuous linear functional (a *distribution*) T_u on $C_c^\infty(\mathbb{R}^2)$ via the pairing

$$\langle T_u, \psi \rangle := \iint_{\mathbb{R}^2} u(x,y)\psi(x,y)\, dA.$$

  Motivated by integration by parts (Green's identities, where boundary terms vanish because ψ is identically zero outside a large ball), the **distributional Laplacian** $\nabla^2 u$ is defined by transferring the Laplace operator $\nabla^2 = \partial_x^2 + \partial_y^2$ onto the test function ψ:

$$\langle \nabla^2 T_u, \psi \rangle := \langle T_u, \nabla^2 \psi \rangle = \iint_{\mathbb{R}^2} u(x,y)\nabla^2\psi(x,y)\, dA.$$

- **Fundamental Solution of the 2D Laplacian:** Even though $u(x,y) = \log\left(\sqrt{x^2+y^2}\right) = \log|\mathbf{x}|$ satisfies $\nabla^2 \log|\mathbf{x}| = 0$ for all $(x,y) \neq (0,0)$ (see Exercise 10.3), its singularity at the origin concentrates mass! Lemma 9.2 proves that in the sense of distributions,

$$\nabla^2 \left(\frac{1}{2\pi} \log \sqrt{x^2+y^2} \right) = \delta_0,$$

  where $\langle \delta_0, \psi \rangle = \psi(0,0)$. That is, $\frac{1}{2\pi}\log|\mathbf{x}|$ is the *fundamental solution* (or Green's function) of the 2D Laplace operator.
- **Connection to Brown Measure of Non-Normal Operators:** For an $n \times n$ matrix $A \in M_n(\mathbb{C})$ with eigenvalues $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, notice that the normalized log-determinant is

$$\frac{1}{n}\log|\det(A - zI)| = \frac{1}{n}\sum_{k=1}^{n} \log|\lambda_k - z|.$$

Taking $\frac{1}{2\pi}\nabla_z^2$ (with $z = x + iy \in \mathbb{C} \cong \mathbb{R}^2$) and applying Lemma 9.2 at each shifted singularity $z = \lambda_k$ yields the **empirical spectral measure** of A:

$$\mu_A := \frac{1}{2\pi}\nabla_z^2\Big(\frac{1}{n}\log|\det(A - zI)|\Big) = \frac{1}{n}\sum_{k=1}^{n}\delta_{\lambda_k}.$$

In an infinite-dimensional von Neumann algebra $\mathcal{M}$ equipped with a faithful normal tracial state τ (a II_1 factor), a non-normal operator $T \in \mathcal{M}$ does not have eigenvalues in the classical sense, nor does the spectral theorem apply. However, using the **Fuglede–Kadison determinant** $\Delta(T - zI) := \exp(\tau(\log|T - zI|))$, L. G. Brown (1986) showed that $z \mapsto \log\Delta(T - zI)$ is a subharmonic function on $\mathbb{C}$, so its distributional Laplacian

$$\mu_T := \frac{1}{2\pi}\nabla_z^2\Big(\log\Delta(T - zI)\Big)$$

defines a unique Borel probability measure on the spectrum $\sigma(T)$, called the **Brown measure** of T. Lemma 9.2 is the exact engine that recovers the point masses of eigenvalues from the logarithmic potential!

Remark 9.1 (Proof Strategy). The idea in the proof below is to excise a small disk $B_{r_0}(\vec{0})$ around the origin so that $\log\Big(\sqrt{x^2+y^2}\Big)$ is smooth on the annulus $A_{r_0,R} = B_R(\vec{0}) \setminus \overline{B_{r_0}(\vec{0})}$, and then use the Generalized Green's Theorem in flux form (4.1b) to reduce the double integral over $A_{r_0,R}$ to two boundary line integrals. The line integral over the outer circle ∂B_R is immediately zero because $\mathrm{supp}(\psi) \subset B_R(\vec{0})$. The line integral over the inner circle ∂B_{r_0} splits into two terms: the first vanishes as $r_0 \to 0^+$ because $\lim_{r_0\to 0^+} r_0\log(r_0) = 0$ (10.1), where the factor r_0 comes from the arc-length element $ds = r_0\,dt$; the second evaluates to $2\pi\psi(0,0)$ by continuity of ψ at the origin.

Recall that $\nabla = \langle\partial_x, \partial_y\rangle$, and taking the formal dot product yields the Laplace operator $\nabla^2 = \nabla\cdot\nabla = \partial_x^2 + \partial_y^2$, where we use the shorthand notation ∂_x and ∂_y for $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$, respectively.

Lemma 9.2 (Fundamental Solution to the Distributional Laplacian)**.** *For any test function $\psi \in C_c^\infty(\mathbb{R}^2)$,*

$$\iint_{\mathbb{R}^2}\log\Big(\sqrt{x^2+y^2}\Big)\nabla^2\psi(x,y)\,dA = 2\pi\iint_{\mathbb{R}^2}\psi(x,y)\,d\delta_0 = 2\pi\cdot\psi(0,0),$$

where $\nabla = \langle\partial_x, \partial_y\rangle$ denotes the del operator in Cartesian coordinates.

Proof. First observe that for any $f, g \in C^2(U)$ on an open set $U \subseteq \mathbb{R}^2$,

$$\nabla\cdot(f\,\nabla g - g\,\nabla f) = f\,\nabla^2 g - g\,\nabla^2 f.^{1} \tag{9.1}$$

This identity (Green's second identity) allows us to apply the divergence version of Generalized Green's Theorem (4.1b) to the vector field $\vec{F} = f\nabla g - g\nabla f$.

Now fix a test function $\psi \in C_c^\infty(\mathbb{R}^2)$. Since ψ has compact support, we can choose $R > 0$ large enough so that $\text{supp}(\psi) \subset B_R(\vec{0}) = \{(x,y) \in \mathbb{R}^2 : \sqrt{x^2+y^2} < R\}$. Consequently, ψ and all of its partial derivatives vanish identically on and outside the outer boundary circle ∂B_R.

Since $\log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi$ is integrable on $B_R(\vec{0})$ (in polar coordinates, $r\log(r)$ is bounded as $r \to 0^+$), we have by the Dominated Convergence Theorem:

$$\iint_{\mathbb{R}^2} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi(x,y)\,dA = \lim_{r_0\to 0^+} \iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi(x,y)\,dA,$$

where $A_{r_0,R} = \{(x,y) \in \mathbb{R}^2 : r_0 \le \sqrt{x^2+y^2} \le R\}$ denotes the annulus centered at the origin with inner radius $r_0 \in (0,R)$ and outer radius R.

On $A_{r_0,R}$ (which excludes the origin), both $f(x,y) = \log\left(\sqrt{x^2+y^2}\right)$ and $g(x,y) = \psi(x,y)$ are smooth (C^∞). Applying (9.1) gives

$$\log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi = \nabla\cdot\left(\log\left(\sqrt{x^2+y^2}\right)\nabla\psi - \psi\,\nabla\log\left(\sqrt{x^2+y^2}\right)\right) + \psi(x,y)\nabla^2\log\left(\sqrt{x^2+y^2}\right).$$

Since $\log\left(\sqrt{x^2+y^2}\right)$ is harmonic on $\mathbb{R}^2 \setminus \{(0,0)\}$ (see Exercise 10.3), we have

$$\psi(x,y)\nabla^2\log\left(\sqrt{x^2+y^2}\right) = 0 \quad \text{for all } (x,y) \in A_{r_0,R}.$$

Therefore, setting

$$\vec{F}(x,y) := \log\left(\sqrt{x^2+y^2}\right)\nabla\psi(x,y) - \psi(x,y)\,\nabla\log\left(\sqrt{x^2+y^2}\right),$$

we can apply the flux form of Generalized Green's Theorem (4.1b) on the annulus $A_{r_0,R}$:

$$\iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi\,dA = \iint_{A_{r_0,R}} \nabla\cdot\vec{F}\,dA \overset{\text{Green}}{=} \oint_{\partial B_{r_0}} \vec{F}\cdot\hat{n}\,ds + \oint_{\partial B_R} \vec{F}\cdot\hat{n}\,ds, \tag{9.2}$$

where $\hat{n}$ denotes the unit normal pointing **outward from the annulus** $A_{r_0,R}$.

By our choice of R, ∂B_R lies strictly outside $\text{supp}(\psi)$, so $\psi \equiv 0$ and $\nabla\psi \equiv \vec{0}$ on ∂B_R. Hence $\vec{F} \equiv \vec{0}$ on ∂B_R, and the second line integral in (9.2) is zero. Thus (9.2) simplifies to

$$\iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi\,dA = \oint_{\partial B_{r_0}} \left(\log\left(\sqrt{x^2+y^2}\right)\nabla\psi - \psi\,\nabla\log\left(\sqrt{x^2+y^2}\right)\right)\cdot\hat{n}\,ds. \tag{9.3}$$

Computation of the Normal Derivatives on ∂B_{r_0}:

[1] This identity uses the product rule for divergence and the cancellation of $(\nabla f)\cdot(\nabla g)$. See Exercise (10.2).

- *Outward unit normal to $A_{r_0,R}$ on ∂B_{r_0}:* Because $A_{r_0,R}$ lies outside the disk $B_{r_0}(\vec{0})$, the normal vector $\hat{n}$ pointing *out of* $A_{r_0,R}$ points **radially inward toward the origin**:

$$\hat{n}(x,y) = -\frac{1}{\sqrt{x^2+y^2}}\langle x,y\rangle = -\frac{1}{r_0}\langle x,y\rangle \quad \text{on } \partial B_{r_0}.$$

- *First term ($\nabla\psi\cdot\hat{n}$):* In polar coordinates $(x,y) = (r\cos\theta, r\sin\theta)$, the multivariable chain rule gives the radial directional derivative

$$\frac{\partial\psi}{\partial r} = \frac{\partial\psi}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial\psi}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial\psi}{\partial x}\cos\theta + \frac{\partial\psi}{\partial y}\sin\theta = \nabla\psi\cdot\frac{\langle x,y\rangle}{\sqrt{x^2+y^2}}.$$

  Therefore, on ∂B_{r_0},

$$\nabla\psi\cdot\hat{n} = -\frac{\partial\psi}{\partial r}.$$

- *Second term ($\nabla\log\left(\sqrt{x^2+y^2}\right)\cdot\hat{n}$):* By the chain rule,

$$\nabla\log\left(\sqrt{x^2+y^2}\right) = \frac{1}{2}\nabla\log\left(x^2+y^2\right) = \left\langle\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}\right\rangle = \frac{1}{x^2+y^2}\langle x,y\rangle.$$

  Taking the dot product with $\hat{n} = -\frac{1}{\sqrt{x^2+y^2}}\langle x,y\rangle$ on ∂B_{r_0} (where $x^2+y^2 = r_0^2$) gives

$$-\psi\,\nabla\log\left(\sqrt{x^2+y^2}\right)\cdot\hat{n} = -\psi\left(\frac{1}{r_0^2}\langle x,y\rangle\right)\cdot\left(-\frac{1}{r_0}\langle x,y\rangle\right) = +\psi\frac{x^2+y^2}{r_0^3} = \frac{1}{r_0}\psi.$$

Substituting these two dot products into (9.3) and factoring out the constants $\log(r_0)$ and $\frac{1}{r_0}$ on the circle ∂B_{r_0} gives

$$\iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi\,dA = -\log(r_0)\oint_{\partial B_{r_0}}\frac{\partial\psi}{\partial r}\,ds + \frac{1}{r_0}\oint_{\partial B_{r_0}}\psi\,ds. \tag{9.4}$$

Next, parameterize the circle ∂B_{r_0} by $\vec{r}(t) = \langle r_0\cos t, r_0\sin t\rangle : [0,2\pi] \to \mathbb{R}^2$. Then $\|\vec{r}\,'(t)\| = r_0$, so the arc-length differential is $ds = \|\vec{r}\,'(t)\|\,dt = r_0\,dt$. Substituting $ds = r_0\,dt$ into (9.4) yields

$$\iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right)\nabla^2\psi\,dA = -r_0\log(r_0)\int_0^{2\pi}\frac{\partial\psi}{\partial r}(\vec{r}(t))\,dt + \int_0^{2\pi}\psi(\vec{r}(t))\,dt. \tag{9.5}$$

Since ψ is continuous on $\mathbb{R}^2$, $\psi(r_0\cos t, r_0\sin t) \to \psi(0,0)$ uniformly for $t \in [0,2\pi]$ as $r_0 \to 0^+$, which justifies bringing the limit inside the second integral:[2]

$$\lim_{r_0\to 0^+}\int_0^{2\pi}\psi(\vec{r}(t))\,dt = \int_0^{2\pi}\psi(0,0)\,dt = 2\pi\,\psi(0,0).$$

[2]Uniform convergence on the compact interval $[0,2\pi]$ (or the Bounded Convergence Theorem, since $|\psi(x,y)| \le \|\psi\|_\infty < \infty$) guarantees $\lim_{r_0\to 0^+}\int_0^{2\pi}\psi(\vec{r}(t))\,dt = \int_0^{2\pi}\lim_{r_0\to 0^+}\psi(\vec{r}(t))\,dt$.

For the first integral, since $\psi \in C_c^\infty(\mathbb{R}^2)$, its gradient $\nabla\psi$ is continuous and hence bounded by some constant $M = \sup_{\mathbb{R}^2} \|\nabla\psi\| < \infty$. Thus $\left|\int_0^{2\pi} \frac{\partial\psi}{\partial r}(\vec{r}(t))\,dt\right| \leq 2\pi M$, and by Exercise (10.1) ($\lim_{r_0\to 0^+} r_0 \log(r_0) = 0$) with the Squeeze Theorem, we obtain

$$\lim_{r_0\to 0^+} \left(-r_0 \log(r_0) \int_0^{2\pi} \frac{\partial\psi}{\partial r}(\vec{r}(t))\,dt\right) = 0.$$

Combining both limits completes the proof:

$$\lim_{r_0\to 0^+} \iint_{A_{r_0,R}} \log\left(\sqrt{x^2+y^2}\right) \nabla^2\psi\, dA = 2\pi\,\psi(0,0). \qquad \square$$

10. Exercises on the Application

Exercise 10.1. Show that

$$\lim_{r\to 0^+} r\log(r) = 0.$$

Hint: Rewrite $r\log(r) = \frac{\log(r)}{1/r}$ and apply L'Hôpital's Rule as $r \to 0^+$:

$$\lim_{r\to 0^+} \frac{\log(r)}{1/r} \overset{\text{L'H}}{=} \lim_{r\to 0^+} \frac{1/r}{-1/r^2} = \lim_{r\to 0^+}(-r) = 0.$$

Exercise 10.2. In the context of Lemma (9.2), use the product rule for derivatives to show that for $f, g \in C^2(\mathbb{R}^2)$,

$$\nabla\cdot(f\,\nabla g) = (\nabla f)\cdot(\nabla g) + f\,\nabla^2 g,$$

and deduce Green's identity $\nabla\cdot(f\,\nabla g - g\,\nabla f) = f\,\nabla^2 g - g\,\nabla^2 f$.
Hint: Recall that $\nabla\cdot(f\,\nabla g) = \langle\partial_x,\partial_y\rangle\cdot\langle f\partial_x g, f\partial_y g\rangle = \partial_x(f\partial_x g)+\partial_y(f\partial_y g)$. Expand each term via the single-variable product rule.

Exercise 10.3. Show that $\log\left(\sqrt{x^2+y^2}\right)$ is harmonic on $\mathbb{R}^2\setminus\{(0,0)\}$. That is, show

$$\partial_x^2\log\left(\sqrt{x^2+y^2}\right) + \partial_y^2\log\left(\sqrt{x^2+y^2}\right) = 0$$

for all $(x,y)\neq(0,0)$.
Hint: Write $\log\left(\sqrt{x^2+y^2}\right) = \frac{1}{2}\log(x^2+y^2)$. Then $\partial_x\left[\frac{1}{2}\log(x^2+y^2)\right] = \frac{x}{x^2+y^2}$, and by the quotient rule,

$$\partial_x^2\log\left(\sqrt{x^2+y^2}\right) = \frac{(x^2+y^2)-x(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}.$$

By symmetry in x and y, $\partial_y^2\log\left(\sqrt{x^2+y^2}\right) = \frac{x^2-y^2}{(x^2+y^2)^2}$, which sums with the x-term to 0.

11. Appendix

Terminology.

(1) **Vector Field:** A map $\vec{F} : D \subseteq \mathbb{R}^n \to \mathbb{R}^n$ that assigns a vector $\vec{F}(\vec{x})$ to each point $\vec{x} \in D$. In $\mathbb{R}^2$, $\vec{F}(x, y) = \langle P(x, y), Q(x, y)\rangle$; in $\mathbb{R}^3$, $\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z)\rangle$.

(2) **Gradient Field / Conservative Field:** A vector field $\vec{F}$ is *conservative* on an open region U if there exists a differentiable scalar function $f : U \to \mathbb{R}$ (called a *scalar potential function*) such that $\vec{F} = \nabla f$. On a simply connected region, a C^1 vector field is conservative if and only if $\nabla \times \vec{F} = \vec{0}$ (or $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$ in $\mathbb{R}^2$).

(3) **Line Integral (over a Scalar Field or Vector Field):**
- *Scalar line integral* along a smooth curve C parameterized by $\vec{r}(t)$, $t \in [a, b]$:
$$\int_C f \, ds = \int_a^b f(\vec{r}(t))\|\vec{r}\,'(t)\| \, dt.$$
- *Vector line integral (Work)*:
$$\int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \hat{T} \, ds = \int_a^b \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t) \, dt.$$

(4) **Surface Integral (over a Scalar Field or Vector Field):**
- *Scalar surface integral* over a surface S parameterized by $\vec{r}(u, v)$, $(u, v) \in D$:
$$\iint_S f \, dS = \iint_D f(\vec{r}(u, v))\|\vec{r}_u \times \vec{r}_v\| \, du \, dv.$$
- *Vector surface integral (Flux)* across an oriented surface S with unit normal $\hat{n} = \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$:
$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n} \, dS = \iint_D \vec{F}(\vec{r}(u, v)) \cdot (\vec{r}_u \times \vec{r}_v) \, du \, dv.$$

(5) **Circulation:** The line integral $\oint_C \vec{F} \cdot \hat{T} \, ds$ of a vector field around a *closed* curve C, measuring the net tendency of the field to circulate along C.

(6) **Flux:** The integral of the normal component $\vec{F} \cdot \hat{n}$ of a vector field across a curve in $\mathbb{R}^2$ ($\int_C \vec{F} \cdot \hat{n} \, ds$) or across an oriented surface in $\mathbb{R}^3$ ($\iint_S \vec{F} \cdot \hat{n} \, dS$), measuring the net rate of flow across the boundary.

(7) **Curl ($\nabla \times \vec{F}$):** For $\vec{F} = \langle P, Q, R\rangle$ in $\mathbb{R}^3$, the vector field measuring infinitesimal rotation:
$$\nabla \times \vec{F} = \left\langle \frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}, \; \frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}, \; \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right\rangle.$$

(8) **Divergence ($\nabla \cdot \vec{F}$):** For $\vec{F} = \langle P, Q, R \rangle$, the scalar field measuring infinitesimal net outward expansion (source/sink density):

$$\nabla \cdot \vec{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}.$$

Differential Operators in Different Coordinate Systems.

Operator	**Coordinates**	**Expression**
Gradient ∇f	Cartesian (x, y, z)	$\frac{\partial f}{\partial x}\hat{e}_x + \frac{\partial f}{\partial y}\hat{e}_y + \frac{\partial f}{\partial z}\hat{e}_z$
	Cylindrical (r, θ, z)	$\frac{\partial f}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{e}_\theta + \frac{\partial f}{\partial z}\hat{e}_z$
	Spherical (ρ, ϕ, θ)	$\frac{\partial f}{\partial \rho}\hat{e}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \phi}\hat{e}_\phi + \frac{1}{\rho \sin\phi}\frac{\partial f}{\partial \theta}\hat{e}_\theta$
Divergence $\nabla \cdot \vec{F}$	Cartesian (x, y, z)	$\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$
	Cylindrical (r, θ, z)	$\frac{1}{r}\frac{\partial (rF_r)}{\partial r} + \frac{1}{r}\frac{\partial F_\theta}{\partial \theta} + \frac{\partial F_z}{\partial z}$
	Spherical (ρ, ϕ, θ)	$\frac{1}{\rho^2}\frac{\partial (\rho^2 F_\rho)}{\partial \rho} + \frac{1}{\rho \sin\phi}\frac{\partial (\sin\phi\, F_\phi)}{\partial \phi} + \frac{1}{\rho \sin\phi}\frac{\partial F_\theta}{\partial \theta}$
Laplacian $\nabla^2 f$	Cartesian (x, y, z)	$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$
	Cylindrical (r, θ, z)	$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial f}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}$
	Spherical (ρ, ϕ, θ)	$\frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2 \frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2 \sin\phi}\frac{\partial}{\partial \phi}\left(\sin\phi \frac{\partial f}{\partial \phi}\right) + \frac{1}{\rho^2 \sin^2\phi}\frac{\partial^2 f}{\partial \theta^2}$

References

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